Bocconi

Bachelor of Science
in Economics and Finance

Empirical Evidence of the Dividend Month Premium in the FTSE STAR Segment:

A Multi-Factor Asset Pricing Approach

Advisor:
Prof. Carlo Ambrogio FAVERO

Bachelor of Science thesis by:

Luca DENNERT
student ID no 3200352

Academic Year 2023/2024

Università Commerciale
Luigi Bocconi

Contents

1 Introduction **3**

2 The FTSE STAR Index and the Italian Dividend Taxation Framework **7**

2.1 The FTSE STAR Index 7

2.2 Admission and Governance Requirements 7

2.3 Sector Structure and Capitalization 8

2.4 Liquidity and Investor Base Structure 10

2.5 The Dividend Taxation System in Italy 12

3 Literature review and hypotheses **13**

3.1 The Dividend Month Premium 13

3.2 Market efficiency 13

3.3 Price pressure 14

3.4 Theoretical models 14

3.5 Hypotheses 17

4 Methodology **18**

5 Results **24**

5.1 Returns 24

5.2 Testing of multi-factor asset pricing models 30

5.3 Returns event study 33

6 Conclusion **37**

6.1 Economic implications 38

6.2 Limitations 38

6.3 Contribution 38

6.4 Future research 39

6.5 Final takeaway 39

References **40**

1. Introduction

Dividends have been shown to significantly affect stock price dynamics, both through their informational content, providing an outlook on the present and future cash flows for investors, and through investor trading behaviors, such as preferences of investors for dividend-paying stocks. Over the years, several studies have analyzed the impact of dividends on price changes, particularly regarding the extent to which their effect on prices cannot be fully explained by traditional asset pricing models. The Dividend Month Premium is now a well-documented abnormal return derived from a market anomaly, since Hartzmark and Solomon (2013) first tested its presence in the U.S. stock market. In their paper, the authors study the price movements of companies in the months when they are predicted to issue a dividend, examining whether these movements are explainable by common risk factors and what the causes might be for the presence of the market anomaly. Their analysis yields an anomaly that is robust to common risk factors and is "consistent with price pressure from dividend-seeking investors" (Hartzmark and Solomon 2013). Furthermore, the effect is stronger in less liquid securities with higher dividend yields, consistent with the demand pressure hypothesis. Other studies, such as those by Chai et al. (2025) and Chae and Koo (2020), find similar results in Australia and South Korea, respectively, while using a similar methodology to the one used by Hartzmark and Solomon. Ainsworth and Nicholson (2015) instead test the presence of the anomaly in multiple countries, finding the most consistent results in France, Germany, and Singapore, in addition to the U.S., while also finding through a cross-country analysis that the performance of the anomaly is influenced by differences in tax regimes.

Hartzmark and Solomon trace the anomaly mainly to two primary explanations. The first relates to the catering theories presented by Baker and Wurgler (2004) and Li and Lie (2006), which hypothesize that companies comply with investor demand for dividends, a demand that may have psychological motivations linked to perceptions of security or

guaranteed income, or institutional drivers. The second cause concerns clientele effects arising from heterogeneous tax treatments. The incentive to buy before the ex for tax purposes pushes prices upward as the ex-date approaches. Some results of their analysis are consistent with both drivers, while other evidence to support such drivers is mixed. The authors also consider a list of alternative explanations, such as the seasonality in returns reported by Heston and Sadka (2004) or the fact that dividend-paying months could coincide with earnings announcement months, meaning that the Dividend Month Premium could be driven in part by the earnings announcement premium (Beaver 1968; Frazzini and Lamont 2008).

This paper aims to verify the presence of the anomaly in Italy and specifically in the FTSE STAR market, an equity segment of the Italian Stock Market that includes mid-cap companies that must comply with stringent requirements in terms of transparency, corporate governance, and liquidity. This paper focuses on this specific market mainly because, as demonstrated by Hartzmark and Solomon, the effect is more prevalent in smaller and less liquid markets, such as the FTSE STAR. In fact, although companies listed in this segment must meet liquidity requirements, such as having 35% free float at IPO and the presence of a liquidity provider, liquidity is not comparable to that of large international markets. Another feature that leads us to choose this market for this analysis is its sector diversification. One might think that the most obvious choice for the present analysis would be the FTSE MIB, the Italian index comprising the 40 largest Italian companies. However, since the latter is overexposed to financial and insurance companies, which account for 35% of the companies listed on this market, the FTSE STAR is a more appropriate segment for this analysis.

Another reason why this paper focuses on the FTSE STAR market is to differentiate from the paper of Ainsworth and Nicholson that tests the presence of the anomaly in Italy and other countries. Their work is on the major Italian exchange and does not find robust results in support of the existence of the anomaly. The reasoning behind choosing the FTSE STAR is that by focusing on a segment characterized by elements such as smaller capitalization and lower liquidity, the analysis might produce more significant

results. At the same time, the Italian market is especially interesting to analyze because of its fiscal difference compared to the U.S. and other major countries. As a matter of fact, in Italy, investors are always subject to a flat 26% substitute tax on dividends and capital gains, which means that there should be no tax clientele effect. Positive results should be attributed to catering effects or alternative explanations.

The methodology used in this paper differs slightly from that used by Hartzmark and Solomon, in that it does not regress stock returns in the months in which a dividend payment is predicted, but instead regresses the returns of the 15 trading days prior to the ex-date. If investors buy a stock on or after the ex-date, they are not entitled to the dividend. In addition, a short portfolio is not constructed because of the short constraints of the FTSE STAR segment. Similarly to Hartzmark and Solomon, the robustness of the anomaly to common risk factors, such as market excess return (MKT–RF), size (SMB), value (HML), profitability (RMW), investment (CMA), and momentum (UMD), is assessed by implementing different asset pricing models. The models used for the analysis are the Capital Asset Pricing Model (CAPM); the Fama–French three-factor model (FF3), the Carhart four-factor model (C4F) and the Fama–French five-factor model (FF5).

The analysis yields high and statistically significant alphas supporting the presence of the anomaly in the FTSE STAR segment, especially when a cutoff is introduced for dividend yields above which to hypothetically invest. Securities with high dividend yields tend to have higher returns in the 15 trading days prior to the ex-date and to have higher and statistically significant alpha. The existence of the anomaly can be attributed to the already mentioned low liquidity; the characteristics of the investor base, as the FTSE STAR segment is characterized by a high presence of domestic retail investors attracted by dividends and the lower presence of international arbitrageurs who, by their nature, tend to neutralize anomalies; a particularly strong signaling effect, as in the Italian context, the payment of a dividend is often interpreted as a sign of solidity and the ability to generate future cash flows probably linked to the widespread family ownership structure; and the already discussed influence of the Earnings Announcement Premium, also identified by Hartzmark and Solomon as a possible cause of the anomaly.

The remainder of the paper is structured as follows. The second chapter describes the Italian context and the FTSE STAR segment. The third chapter reviews the literature and discusses the main hypotheses. The fourth chapter presents the methodology used. The fifth chapter discusses the results, and the sixth chapter concludes.

2. The FTSE STAR Index and the Italian Dividend Taxation Framework

2.1 The FTSE STAR Index

The FTSE STAR Index was created in 2001 and is an index of the STAR segment of the Euronext Milan market managed by Borsa Italiana. It comprises 63 medium-sized stocks (August 2025). STAR is an acronym for 'segmento titoli con alti requisiti' (segment for stocks with high requirements). These requirements relate to liquidity, transparency, and corporate governance. The segment currently represents approximately 3.5% of the Italian stock market capitalization. (Borsa Italiana S.p.A. 2025a)

2.2 Admission and Governance Requirements

As noted earlier, the Borsa Italiana, the Italian stock exchange, imposes requirements to join and remain in the segment. This is mainly to guarantee the quality of companies to investors.

In terms of liquidity, companies must have a free float of at least 35% of their share capital in the case of a new listing and at least 20% to remain in the segment. In addition, liquidity providers must be present and maintain simultaneous bid-ask quotes during trading sessions, respecting maximum spread limits and minimum quantities for each side. However, it should be noted that the liquidity requirements of this segment may not be sufficient to make a mid-cap market segment as liquid as large-cap indices and markets such

as the NASDAQ, the S&P 500, or, in Italy, the FTSE MIB. From a transparency perspective, there are several requirements for companies, such as more stringent disclosure obligations than in the ordinary market, committing to provide more information to investors and ensuring visibility and continuity in disclosure, regular publication of financial results and corporate events, and the mandatory formal appointment of an Investor Relations Manager.

Finally, in terms of corporate governance, an adequate number of independent directors and the adequacy of control bodies are required, as well as the adoption of the Corporate Governance Code and the adoption of internal committees (risk control/internal audit, remuneration, appointments), and internal control and risk management systems.

2.3 Sector Structure and Capitalization

From a sector composition perspective, the FTSE STAR segment appears to be well diversified, especially considering that the most represented sector in the segment is Industrials, which is very heterogeneous internally and includes several microsectors that often do not have many characteristics in common. A clear example is the comparison between Avio and Sabaf, both categorized as companies operating in the industrial sector. While the former is "a company in the space propulsion sector, launching institutional, government, and commercial payloads into Earth orbit using the Vega family of rockets" (Avio S.p.A. 2025), the latter is "a manufacturer of components for domestic gas and induction cooking appliances, mainly producing taps, thermostats, gas burners, and components for induction cooking" (Sabaf S.p.A. 2025). In addition to the obvious difference in production, the financial data also differ, with Avio having an average EBITDA margin of 6.6% over the last five years (2020-2024) and Sabaf 15.2% and, respectively, a Beta of 0.58 and 1.06, computed monthly over the last 5 years (Yahoo Finance 2025). Overall, the industrial sector accounts for 27% of the index in terms of number of stocks, demonstrating the strong influence in Italy of industrial and manufacturing SMEs, which are often family-run. The other relevant sectors within the segment are Consumer Goods,

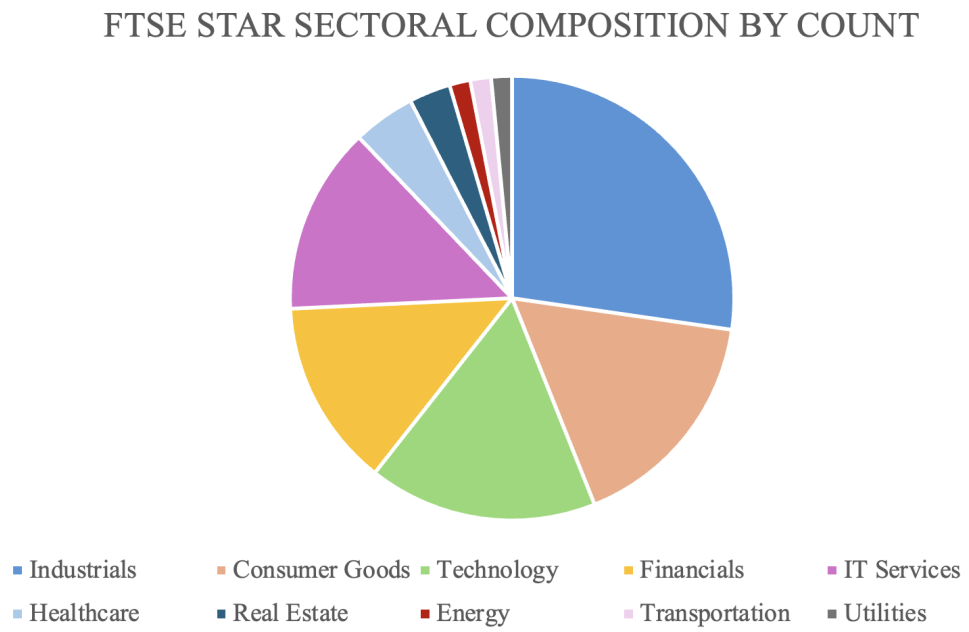


Figure 2.1: Source: Author's calculations based on data from Borsa Italiana, FTSE Russell (ICB), and company documents (latest available financial statements). Note: the FTSE Italia STAR universe is captured as of April 2025.

Technology, Financials, and IT Services, which, however, all account for between 14% and 17% of the total number of companies on the market. It should be noted that the percentages are calculated based on the total number of companies, since the econometric analysis weights the various stocks equally and not according to market value.

It is interesting to note that the FTSE MIB is much more concentrated, especially with regard to the weight of financial and insurance companies compared to the total number of companies in the index. The financial sector accounts for 35% of the total number of companies on the FTSE MIB, which includes Unicredit, Intesa Sanpaolo, Mediobanca, and Generali, among others. By comparison, the most represented sector in the S&P 500 is industrials, which, as we have said, is diversified by nature, accounting for 21% of the total number of companies included in the index, while the financial sector accounts for only about 14% of the total number (Spencer Stuart 2024). This comparison explains one of the reasons why this paper chose to focus on the FTSE STAR.

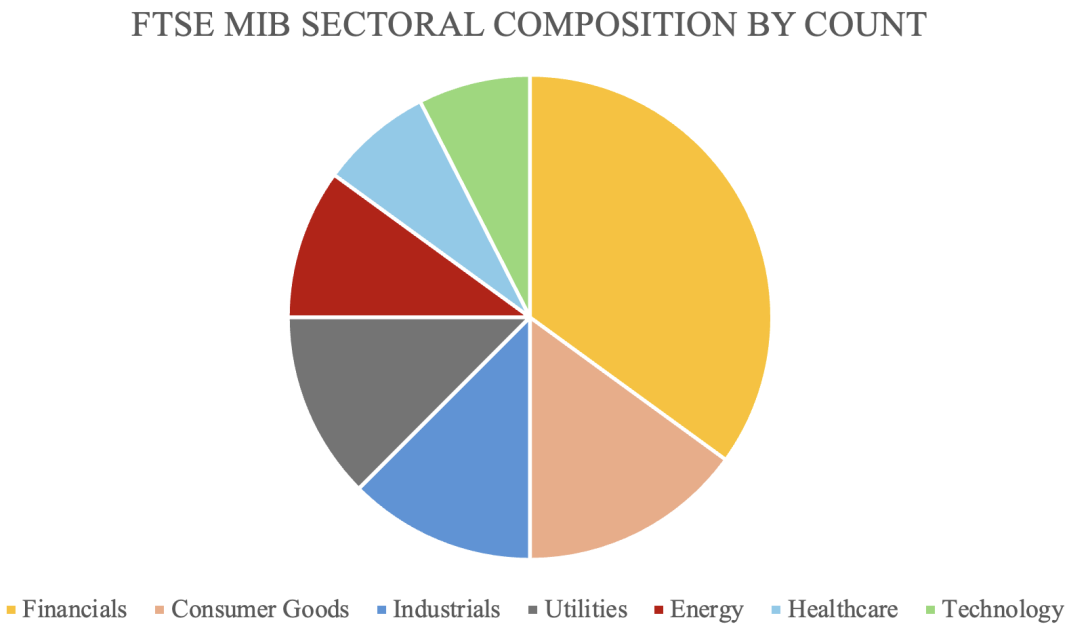


Figure 2.2: Source: Author's calculations based on data from Borsa Italiana and company documents (latest available financial statements). Note: the "FTSE MIB" universe is captured as of April 2025.

2.4 Liquidity and Investor Base Structure

As already mentioned, companies included in the FTSE STAR segment must meet certain liquidity requirements, but this does not necessarily mean that the segment can be defined as highly liquid. In fact, if we compare the requirements of the FTSE STAR with those of the FTSE MIB or U.S. large-cap markets, it is clear that those of the STAR segment are much less stringent in terms of liquidity. Although the liquidity of each STAR segment must be monitored by a specialist, its shares must comply with a maximum spread of 3%, with minimum quantities of at least €2,500 per side and bid-ask presence for at least 60% of the trading session (Borsa Italiana S.p.A. 2025b). The maximum spread of 3% alone indicates that the returns of the strategy analyzed in this paper may be significantly reduced. To provide some context, for FTSE MIB securities where a specialist operates, the parameters are more stringent: maximum spread of 0.5% and minimum quantity of €10,000. This means that the maximum spread for the FTSE STAR can be up to six times higher than the maximum spread for the FTSE MIB. In U.S. large-cap markets, typical spreads are in the single digits of basis points. As reported by Virtu Virtu (2024), the

median spread for the S&P 500 stands at 4.4 bps compared to 18.7 bps for the small caps of the Russell 2000. The analysis carried out in this study using data from LSEG/Refinitiv found that the median spread for the FTSE STAR in the period January 2020-December 2024 stands at 104.4 bps, while the average spread is 160.9, with an average daily turnover of €1,188,493.39. These figures show that the FTSE STAR market is not even remotely comparable to the U.S. small-cap markets in terms of liquidity.

It is also interesting to note the structure of the investor base in Italy, which is markedly heterogeneous in terms of issuer size. According to CONSOB's 2024 report, only 11.2% of the capital of SMEs listed on Euronext Milan is held by institutional investors, while they hold 24.1% of non-SMEs and as much as 31% of large caps (CONSOB – Commissione Nazionale per le Società e la Borsa 2025). Since STAR groups together high-quality mid-small cap companies, it is reasonable to assume that there is less penetration by domestic and international institutional investors, with a relatively greater weighting of domestic retail investors, which increases the likelihood of temporary inefficiencies and, consequently, the emergence of anomalies. Furthermore, as demonstrated by Graham and Kumar (2006), retail investors show a marked preference for dividend yield with age, similar to Becker et al. (2009), who suggest that local demand for dividends is higher where demographics suggest it. This can be applied especially to a country such as Italy, whose demographics are heavily skewed towards older age groups. Additionally, a strong propensity for dividends and coupons on the part of Italian retail investors can be inferred from the "2024 Report on the Investment Choices of Italian Households" published by CONSOB – Commissione Nazionale per le Società e la Borsa (2024). This report indicates that the most frequent investment objectives of Italian households are capital protection (81%), realization of personal projects (70%); attention to expected returns is among the main drivers (30%). The average composition of retail portfolios is as follows: certificates of deposit/postal bonds 48%, government bonds 39%, mutual funds 36%, corporate bonds 35%, term deposits 32%, and stocks 32%. Not only are the investment objectives consistent with the search for stable flows (coupons/dividends), but the strong presence of coupon instruments indicates a preference for periodic income.

2.5 The Dividend Taxation System in Italy

The Italian tax system applies a flat rate of 26% on dividends and capital gains, while for companies resident in Italy, dividends received from Italian or non-blacklisted companies are 95% exempt for IRES (corporate income tax), which means that only 5% contributes to the company's taxable income, and, as a result, dividends are virtually neutral. For non-residents, dividends from Italian companies are also subject to WHT of 26%, but this can be reduced by treaty to a total exemption of 0% if the requirements of the "Parent-Subsidiary" directive for EU companies are met. As for funds, they are exempt from IRES, and taxation typically occurs at 26% on the investor when they receive the income (PwC 2025). This makes the fund fiscally neutral in the choice between dividends and capital gains.

3. Literature review and hypotheses

3.1 The Dividend Month Premium

The Dividend Month Premium tested by Hartzmark and Solomon in their paper consists of the anomalous return differential of a portfolio that goes long on all stocks in the months when they are predicted to distribute dividends and short on otherwise comparable non-dividend-month stocks. Essentially, the premium consists of the alpha of the long-short portfolio relative to standard factor models. In their paper, the authors find an average monthly return of 1.38% for the long portfolio, annualized at 16.56%, and a monthly alpha ranging from 0.404% to 0.472% depending on the model, annualized at 4.848% and 5.664%, all significant at the 1% level.

Significantly positive results were also found by Daniel Chai, Viet Do, and Maria Strydom in Australia, Bonha Koo and Joon Chae in South Korea, and Andrew Ainsworth and Maximilian Nicholson in various countries around the world, but with the most relevant results found in France, Germany, Singapore, and the U.S..

3.2 Market efficiency

As in Hartzmark and Solomon, the null hypothesis is Eugene Fama's semi-strong EMH (Fama 1970): prices reflect all public information, so a trading rule based on known ex-dates should not generate abnormal risk-adjusted returns net of transaction costs. Under this null hypothesis, the Dividend Month Premium should not exist. However, it is well established that anomalies can persist if there are limits to arbitrage. In 1970, Fama listed the absence of transaction costs as one of the three elements sufficient to make a market efficient. High transaction costs can affect market efficiency as they could prevent

arbitrageurs from eliminating market anomalies, erasing the profits that arbitrageurs might realize. We have already seen how the FTSE STAR is a generally illiquid market with relatively high spreads that represent implicit transaction costs that could cause the anomaly to persist.

We evaluate abnormal returns relative to several asset-pricing benchmarks (CAPM, FF3, Carhart, FF5). Due to the joint-hypothesis problem, significant alphas indicate deviations conditional on the chosen model and do not, by themselves, establish a violation of market efficiency.

3.3 Price pressure

The main alternative hypothesis, as identified by Hartzmark and Solomon, is that returns are high in the days prior to the ex-date due to price pressures resulting from trading by investors on the dividends themselves and by liquidity providers and arbitrageurs hoping to profit from short-term price movements, which determine the structure of supply and demand for dividend-paying securities. Hartzmark and Solomon hypothesize that if investors receive utility from dividends and interest rates are positive, these investors will be willing to pay more than others to receive those dividends at a short distance. In their paper, the authors also hypothesize that trades by arbitrageurs and liquidity providers may not be able to offset the trades of investors particularly attracted to dividends, which could lead to upward pressure on prices before the ex-date and a reversal thereafter.

3.4 Theoretical models

As already mentioned, among the main causes attributable to the anomaly are catering theories, such as those presented by Baker and Wurgler (2004), according to which, for psychological reasons referred to investors or institutional reasons, companies comply with investors' requests regarding dividends. As we analyzed in the previous

chapter, this theory could perfectly be consistent with what we observe in the FTSE STAR segment, where the presence of institutional and international investors is very low and where retail investors dominate. It is precisely this category that in Italy is extremely attracted to constant and secure income, such as dividends. This can also be linked to the paper by Shefrin and Statman (1984), which argues that the preference for dividends is explained by investors' goals of self-control and mental accounting. It can be assumed that these biases are attributable to investors who are less educated in financial matters, given that they are unable to use the most rational investment strategies possible, which would consist of neutrality between dividend income and capital gains, or even a preference by investors for non-dividend-paying securities so as not to have to pay frequent taxes on dividend income and to be able to make the most of the potential of compound interest. It is no coincidence that the Italian average financial literacy score is 53 out of 100, significantly below the OECD average that is 63, according to the International Survey of Adult Financial Literacy (OECD/INFE 2023).

The other main theoretical model identified by Hartzmark and Solomon concerns dividend clientele theories. Investors with different tax treatments may have different preferences in terms of dividends. Michaely et al. (1996) argue in their paper that "The difference in agents' tax status induces them to trade dynamically to minimize their tax liabilities." Similarly, Elton and Gruber (1970) argue that the ex-dividend drop is lower than the dividend due to differences in taxation, which is classic evidence of tax-induced clientele, although the evidence shifts with regulatory changes.

In the United States, the tax treatment of dividends varies depending on the type of investor and dividend. For individuals, there is a difference between qualified and ordinary dividends. The former are taxed at long-term capital gains rates if certain requirements are met, while the latter are taxed at the taxpayer's ordinary rate, which can clearly create differences in the demand for dividends among different investors. In addition, pension funds and endowments do not pay current tax on dividends, and taxation occurs (if at all) only at the time of distribution from the account.

In Australia, the dividend clientele effect can be particularly relevant because of

its unique tax system for dividends. In Australia, there is a difference between franked dividends and unfranked dividends, because until 1997, the former included a tax credit that investors could use to offset their taxes. It is not surprising that until 1997, the Dividend Month Premium was very significant for securities with franked dividends. In 1997, a 45-day holding period was introduced to claim the tax credit, and as a result, the anomaly diminished for this type of security.

As for Italy, the system is characterized by the aforementioned flat substitute tax of 26%, equal to that on capital gains, which should not make one type of income preferable to another for tax reasons. Hence, it would not be logical to attribute the hypothetical existence of the anomaly to tax causes such as those suggested by Michaely, Vila, and Wang.

Hartzmark and Solomon also suggest that the Dividend Month Premium probably derives mainly from catering effects, showing that the DMP is higher than 15 bps during recessions and appears higher in times of high volatility. This would be consistent with the interpretation that dividends are perceived as a safe and guaranteed source of income and the category of stocks that distribute them as a safe haven in times of market uncertainty. At the same time, however, if aggregate liquidity declines during periods of recession or market volatility, trading by tax-related clienteles should also have a greater impact on prices, so it is not entirely clear whether the evidence supporting the first thesis is sufficient.

Another theoretical model consistent with the FTSE STAR case but not discussed by Hartzmark and Solomon is signaling theory. This theory argues that managers use dividends as a costly signal of information about cash flow and future profitability prospects. The signal is credible precisely because distributing dividends is challenging and only solid companies can maintain them over time, as indicated by the classic models of Bhattacharya (1979) and Miller and Rock (1985). In Italy and in the context of the dividend anomaly in the FTSE STAR, these models are particularly consistent given the high information asymmetries of mid-caps compared to blue chips, which boast extensive analyst coverage. In the context of Italian mid-caps, therefore, a stable and growing dividend can serve as a

credible signal. Furthermore, given the aforementioned widespread family or concentrated ownership structure, this raises agency problems between controlling shareholders and minority shareholders. Mancinelli and Ozkan (2006) link the ownership structure of Italian companies to payouts, showing how dividends can be used strategically to credibly signal a commitment to external parties or minority investors to protect their interests as well.

Hartzmark and Solomon also present several possible alternative explanations, such as the fact that the DMP could pick up part of the earnings announcement premium. If the month of the dividend payment is the same as the earnings announcement, the latter's premium could influence the DMP. Another cause could be identified in the seasonality of returns, as identified by Heston and Sadka (2004). However, the authors rule out that these explanations could have an impact on the DMP by testing them.

3.5 Hypotheses

Before delving into the explanation of the methodology used, it is appropriate to list and briefly explain the hypotheses that this paper aims to test. The bottom line is to test whether a dividend market anomaly can be identified on the FTSE STAR segment by looking for alphas that are robust to standard asset pricing models (CAPM, FF3, C4F, FF5). We also want to verify the average return of the 15-day strategy that seeks to exploit the anomaly in the run-up to the ex-date and thus verify the presence of positive average returns before the ex-date and a partial reversal afterwards. In addition, this paper aims to test the liquidity of the STAR segment to verify that the strategy has sufficient margins to support transaction costs and spreads. Finally, we want to test whether the intensity of the pre-run increases with higher dividend yields.

4. Methodology

As already mentioned, the methodology used in this paper differs in part from that used in the paper by Hartzmark and Solomon. In this study, the ex-date is known before the operational window (post-announcement) and the analysis is not based on predicted ex-dates. Data on prices, dividends, spreads, and turnover for companies in the FTSE STAR segment were obtained from LSEG/Refinitiv, while data on factors were obtained from Kenneth French's data library. The analysis focuses on the period from January 2020 to December 2024. In total during the 5 years, companies from the FTSE STAR paid dividends 214 times, with an average net dividend yield of 2.44%.

First, we wanted to calculate the average returns of the strategy in the 15 trading days prior to the ex-date by computing the returns in the 15-day window for each observation (formula 4.1) and then averaging them and dividing them into cutoff brackets for dividend yield, computed as net dividend yield divided by the price at t-15, where t is the ex-date. We set the baseline window at 15 trading days before the ex-date [t-15,t-1] for three reasons. i) The pre-ex-date run-up seems to be concentrated in this window. ii) Shorter windows (e.g., 5/10 days) risk truncating the positioning phase, while longer windows ($\geq$ 20 days) increase contamination from unrelated news (earnings, guidance, macro releases) and raise overlap across events, reducing independence. (iii) Operationally, a 15-day horizon is implementable and consistent with standard event-study practice around corporate actions. We also choose to use the dividend yield calculated as $D/P_{i,T-15}$ and therefore with the price at t-15, to keep the signal ex-ante and avoid mechanical effects deriving from the run-up on P. We close at t-1 to exclude the dividend and the mechanical drop in price at t = 0. We denote the ex-date by T and calendar/event time by t, with $t = 0$ at the ex-date and $t \in [-15, +15]$. We then verified whether liquidity could affect the average returns of the strategy by calculating the average and median spreads and turnover. We also performed the same analysis on average returns but taking spreads into

account and therefore buying at t-15 at the ask price and selling at t-1 at the bid price (formula 4.2).

For the model testing, European factors were used for this paper. The reason for this choice is that local or regional models outperform global models. Griffin (2002) demonstrates how country-specific models explain more variance and have lower pricing errors than models with global factors. World factors suffer from integration heterogeneity and currency mix, as partly demonstrated by Fama and French (Fama and French 1993)/ (2015). At the same time, however, it is not optimal to construct Italian factors, even though it would be ideal in principle to use perfectly optimal factors. This is because the Italian universe, especially the FTSE STAR segment, is small and because there are no long, standardized series shared as academic benchmarks. In practice, French's regional factors offer a balance between local adherence and statistical robustness.

To arrive at the alphas, the cumulative returns for the 15 days prior to the ex-date were first calculated for each dividend observation for each company in the FTSE STAR segment using formula 4.1. The cumulative factor values were then calculated in the window of each observation using formula 4.3.

$$\mathrm{CR}_{i,T} = \frac{P_{i,T-1} - P_{i,T-15}}{P_{i,T-15}} \tag{4.1}$$

$\mathrm{CR}_{i,T}$: cumulative return of stock i computed over the 15 periods preceding the ex-date T in the window $[T-15,\, T-1]$
i: i-th stock, i= 1, . . . , n
$P_{i,t}$: closing price of stock i at period t
T: ex-date, t=0

$$\mathrm{CAR}_{i,T} = \frac{\mathrm{Bid}_{i,T-1} - \mathrm{Ask}_{i,T-15}}{\mathrm{Ask}_{i,T-15}} \tag{4.2}$$

$\mathrm{CAR}_{i,T}$: cumulative adjusted return of stock i computed over the 15 periods preceding the ex-date T in the window $[T-15,\, T-1]$
i: i-th stock, i= 1, . . . , n
$Bid_{i,T-1}$: closing bid price of stock i at $T-1$
T: ex-date

$$\mathrm{CFR}_{k,T} = \left(\prod_{t=T-15}^{T-1} (1 + F_{k,t}) \right) - 1 \tag{4.3}$$

$\mathrm{CFR}_{k,T}$: cumulative return of factor k computed over the 15 periods preceding the ex-date T in the window $[T-15,\, T-1]$
k: risk factor index (MKT, SMB, HML, RMW, CMA, UMD)
$F_{k,t}$: periodic return of factor k at period t (expressed in decimals)
T: ex-date

To test the robustness of the risk factor anomaly, it was decided to test the cumulative returns described above with standard asset pricing models. The first model used is the well-known Capital Asset Pricing Model (CAPM) proposed by Sharpe (1964) and represented by formula 4.4. Using this model, Sharpe hypothesizes that the excess return of a security is entirely explained by its exposure to systematic risk measured by the company's specific beta, based on the mean-variance preferences derived from the Neumann and Morgenstern (1947) utility theorem. According to CAPM, a company's idiosyncratic risk should not affect its valuation, since idiosyncratic risk is virtually eliminated if the investor rationally diversifies the portfolio (Markowitz 1952)/(1959). Many studies have questioned the validity of the model and shown how the presence of alphas conflicts with the original theoretical model (Jensen 1968; Black et al. 1972; Fama and MacBeth 1973). To assess whether the pre-ex-date returns contain abnormal performance beyond systematic risk, Jensen-style alphas are estimated using the index model, shown in formula 4.5. A statistically positive alpha indicates an anomaly or risk-adjusted excess return consistent with a pre-ex-date premium.

$$E(r_i) = r_f + \beta_i \left(E(r_M) - r_f \right) \tag{4.4}$$

$E(r_i)$: expected return of i-th security/portfolio
r_f: risk-free rate
β_i: systematic risk beta/loading of i-th security/portfolio
$E(r_M)$: expected market return

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_i \left(r_{M,t} - r_{f,t} \right) + \varepsilon_{i,t} \tag{4.5}$$

$r_{i,t}$: realized return of i-th security/portfolio at time t
$r_{f,t}$: risk-free rate at time t
α_i: intercept/abnormal return of i-th security/portfolio
$r_{M,t}$: realized market return at time t
$\varepsilon_{i,t}$: error term of i-th security/portfolio at time t

In addition, two of the models hypothesized by Fama and French (1993)/ (2015) were tested with the aim of finding models with greater explanatory power than that of the CAPM, which explains on average 70% of the returns of diversified portfolios within the sample (Foye et al. 2013). The first model developed in 1993 is the Fama–French three-factor model (FF3), which adds two new factors to the CAPM: size (SMB) and value (HML). The size factor represents the outperformance of small companies over large companies in terms of capitalization, while the value factor represents the outperformance of companies with high book/market ratios over companies with low book/market ratios. The FF3 model exceeds on average 90% explanatory power, and the empirical model is represented by formula 4.6. In 2015, Fama and French added the factors profitability (RMW) and investments (CMA) to the FF3, defined similarly to the SMB and HML factors. Respectively, they are the difference between the returns of firms with robust and weak operating profitability and the difference between the returns of companies that invest conservatively and companies that invest aggressively. The model is represented in formula 4.7.

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,\mathrm{Mkt-rf}}(r_{M,t} - r_{f,t}) + \beta_{i,\mathrm{SMB}}\,\mathrm{SMB}_t + \beta_{i,\mathrm{HML}}\,\mathrm{HML}_t + \varepsilon_{i,t} \tag{4.6}$$

$r_{i,t}$: realized return of i-th security/portfolio at time t
α_i: intercept/abnormal return of i-th security/portfolio
$r_{f,t}$: risk-free rate at time t
$\beta_{i,\mathrm{Mkt-rf}}$: systematic risk beta/loading of i-th security/portfolio
$\beta_{i,\mathrm{SMB}}$: small-minus-big beta/loading of i-th security or portfolio
$\beta_{i,\mathrm{HML}}$: high-minus-low beta/loading of i-th security/portfolio
SMB_t: small-minus-big factor at time t
HML_t: high-minus-low factor at time t
$\varepsilon_{i,t}$: error term of i-th security/portfolio at time t

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,\text{Mkt-rf}}(r_{M,t} - r_{f,t}) + \beta_{i,\text{SMB}}\,\text{SMB}_t + \beta_{i,\text{HML}}\,\text{HML}_t + \beta_{i,\text{RMW}}\,\text{RMW}_t + \beta_{i,\text{CMA}}\,\text{CMA}_t + \varepsilon_{i,t} \tag{4.7}$$

$\beta_{i,\text{RMW}}$: robust-minus-weak beta/loading of i-th security or portfolio
$\beta_{i,\text{CMA}}$: conservative-minus-aggressive beta/loading of i-th security/portfolio
RMW_t: robust-minus-weak factor at time t
CMA_t: conservative-minus-aggressive factor at time t

In 1997, Carhart proposed adding a fourth factor, momentum (UMD), to the Fama–French three-factor model to consider the existence of outperforming and underperforming securities (Carhart 1997). The Carhart 4-factor model (C4F) is the latest model used in this paper and is represented by formula 4.8.

$$r_{i,t} - r_{f,t} = \alpha_i + \beta_{i,\text{Mkt-rf}}(r_{M,t} - r_{f,t}) + \beta_{i,\text{SMB}}\,\text{SMB}_t + \beta_{i,\text{HML}}\,\text{HML}_t + \beta_{i,\text{UMD}}\,\text{UMD}_t + \varepsilon_{i,t} \tag{4.8}$$

$\beta_{i,\text{UMD}}$: up-minus-down beta/loading of i-th security/portfolio
UMD_t: up-minus-down factor at time t

$$CR_{i,T} - CRF_{f,T} = \alpha_i + \sum_{k=1}^{n} \beta_{i,\text{k}} CFR_{k,T} \tag{4.9}$$

$CRF_{f,T}$: cumulative risk free rate defined as $\left(\prod_{t=T-15}^{T-1}(1 + r_{f,t})\right) - 1$
$\sum_{k=1}^{n} \beta_{i,\text{k}} CFR_{k,T}$: sum of products between security/portfolio i-th beta for factor k and cumulative return of factor k at time T; the number and which factors k are used vary depending on the model
Note: to perform the same analysis but taking into account bid-ask spreads, $CR_{i,T}$ must be replaced with $CAR_{i,T}$

The cumulative returns for each observation calculated using formula 4.1, minus the risk-free rate ($CR_{i,T} - r_{f,T}$), were then regressed on the cumulative factors of formula

4.3 associated with each observation based on the model used (formula 4.9). The purpose of these regressions is to verify the existence of positive and statistically significant alphas.

The same analysis was then performed, but with the addition of various dividend yield brackets as cutoffs. The rationale is to verify whether a portfolio that invests only in companies with a dividend yield above a certain threshold for the 15 trading days prior to the ex-date has better returns and higher, statistically significant alphas. For completeness, all brackets from 0% to 5.5% were analyzed with increments of 0.25%.

Finally, the performance of the strategy's returns around the ex-date was analyzed, also to verify a hypothetical post-ex-date reversal, computing the cross-sectional average return for each event time $t \in [-15, +15]$.

5. Results

5.1 Returns

The results relating to the strategy's returns expressed in terms of the reference window, i.e., the 15 days prior to the ex-date, are shown in Table 5.1, while those in annualized terms are shown in Table 5.2, which includes both linearly annualized and compound annualized returns.

The average returns are all positive regardless of the cutoff. Furthermore, returns tend to increase as the cutoff for dividend yield increases, with the minimum return of 1.666% corresponding to the cutoff of 0.25% and the maximum return of 6.424% corresponding to the cutoff of 4.5%. All dividend yield cutoffs show statistically positive average returns over 15 days (right-tailed t-test). In most cases, the p-value is ≤ 0.01; for buckets with higher yields ($\geq 5.25\%$), where the number of observations is lower, the p-value is instead < 0.05. However, it is important to note that the samples for strategies with high cutoffs are extremely small. In fact, all strategies with cutoffs equal to or greater than 4.25% have samples of fewer than 30. The standard deviation is in the range [7.663, 10.774], while the Sharpe ratio is in the range [0.186, 0.622]. Consistent with the hypothesis regarding price pressure resulting from the cash dividend preferences of retail investors characteristic of the FTSE STAR segment, the premium increases with the dividend yield. Furthermore, as the cutoff increases, the average increases more than the volatility, and the Sharpe ratio rises: the demand is more price-inelastic on high-yield names. This is also consistent with the price pressure of dividend-seeking investors. The annualized data are also interesting in order to make the results more easily comparable with other strategies, although it should be noted that annualization assumes that returns are independent and identically distributed and that windows do not overlap, with seasonality and clustering

potentially inflating annualizations. The strategy is not feasible throughout the year given the small number of companies and, consequently, dividends distributed during the year. Linearly annualized returns are very high, ranging from 27.989% to 107.293%, while the range for compound annualized returns is [31.993%, 184.62]. Although not very significant, it is interesting to compare this, for illustrative purposes only, this with the S&P 500, which from 1928 to the first quarter of 2025 averaged an annual return of 9.96%. Figure 5.3 also includes the values of the means ± 1 and 2 standard deviations for each dividend yield. It is important to note, however, that the Jarque-Bera test rejects the null hypothesis of normal distribution of returns, with a p-value of 1.53E-16. The empirical distribution is shown in figure 5.4.

Cutoff DivYield (%)	**N events**	**AvgReturn 15d (%)**	**StdDev 15d (%)**	**Sharpe 15d**
0	214	1.680***	8.146	0.187
0.25	209	1.666***	8.139	0.186
0.5	196	2.025***	7.958	0.235
0.75	175	1.930***	7.844	0.226
1	160	1.957***	7.859	0.228
1.25	145	2.012***	7.663	0.241
1.5	132	2.322***	7.739	0.278
1.75	121	2.703***	7.666	0.329
2	100	3.050***	7.885	0.363
2.25	90	2.872***	7.767	0.347
2.5	82	3.007***	7.880	0.358
2.75	71	3.384***	8.170	0.390
3	63	3.982***	8.427	0.449
3.25	51	4.312***	8.868	0.463
3.5	45	4.544***	8.866	0.490
3.75	39	5.179***	9.291	0.536
4	35	4.982***	9.600	0.499
4.25	28	6.251***	9.815	0.617
4.5	23	6.424***	10.060	0.622
4.75	20	5.567***	9.123	0.592
5	18	5.955***	9.556	0.603
5.25	14	5.655**	10.774	0.509
5.5	14	5.655**	10.774	0.509

Table 5.1: Returns summary (15 days window)

Note: *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$

Cutoff DivYield (%)	N events	AvgReturn (lin. ann.,%	AvgReturn (comp. ann.,%)	StdDev 15d (ann.,%))	Sharpe (ann.)
0	214	28.224	32.299	33.389	0.766
0.25	209	27.989	31.993	33.360	0.762
0.5	196	34.020	40.046	32.618	0.963
0.75	175	32.424	37.871	32.151	0.926
1	160	32.878	38.486	32.212	0.935
1.25	145	33.802	39.747	31.409	0.988
1.5	132	39.010	47.055	31.720	1.139
1.75	121	45.410	56.530	31.421	1.348
2	100	51.240	65.656	32.319	1.488
2.25	90	48.250	60.914	31.835	1.422
2.5	82	50.518	64.498	32.298	1.467
2.75	71	56.851	74.911	33.487	1.599
3	63	66.898	92.707	34.540	1.840
3.25	51	72.442	103.243	36.348	1.898
3.5	45	76.339	110.972	36.340	2.008
3.75	39	87.007	133.565	38.082	2.197
4	35	83.698	126.323	39.348	2.045
4.25	28	105.017	176.946	40.230	2.529
4.5	23	107.923	184.620	41.234	2.549
4.75	20	93.526	148.470	37.393	2.426
5	18	100.044	164.265	39.168	2.472
5.25	14	95.004	151.972	44.160	2.086
5.5	14	95.004	151.972	44.160	2.086

Table 5.2: Returns summary (annualized)

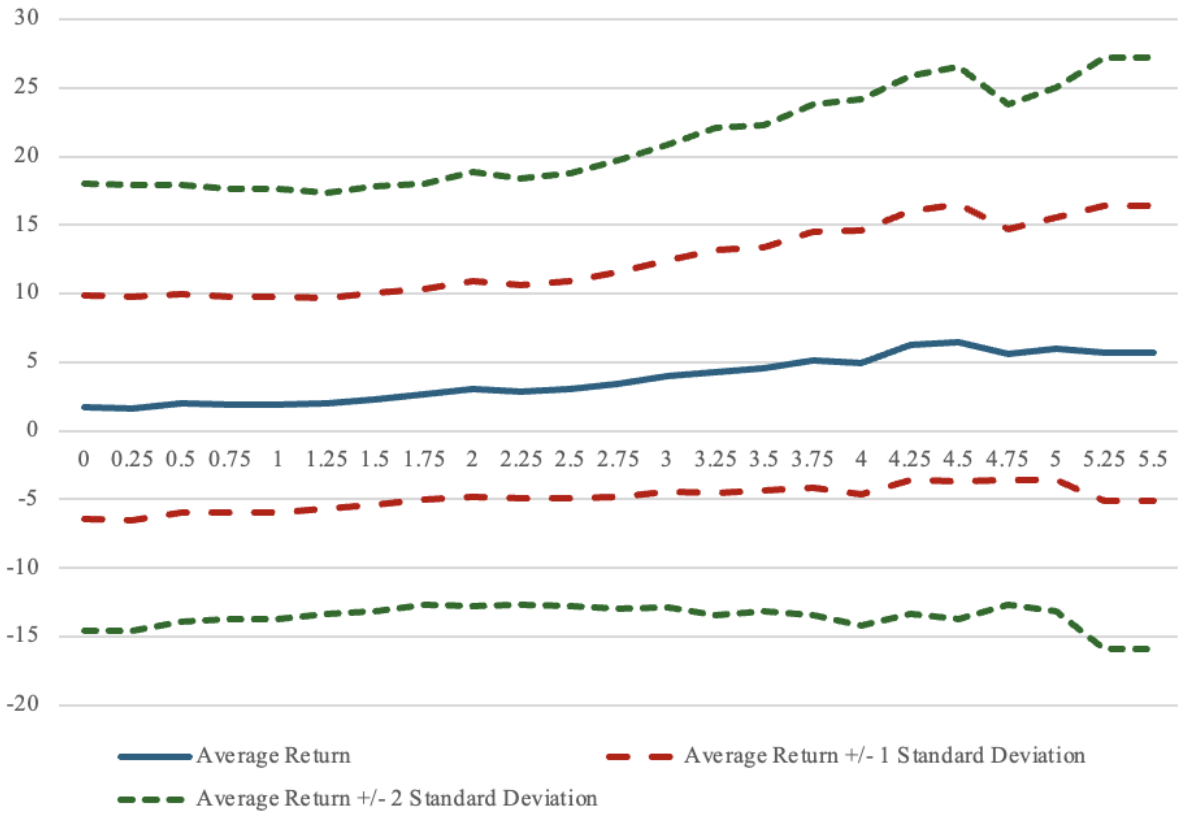


Figure 5.3: Average returns for 15 days ± 1 & 2 standard deviations (in %)

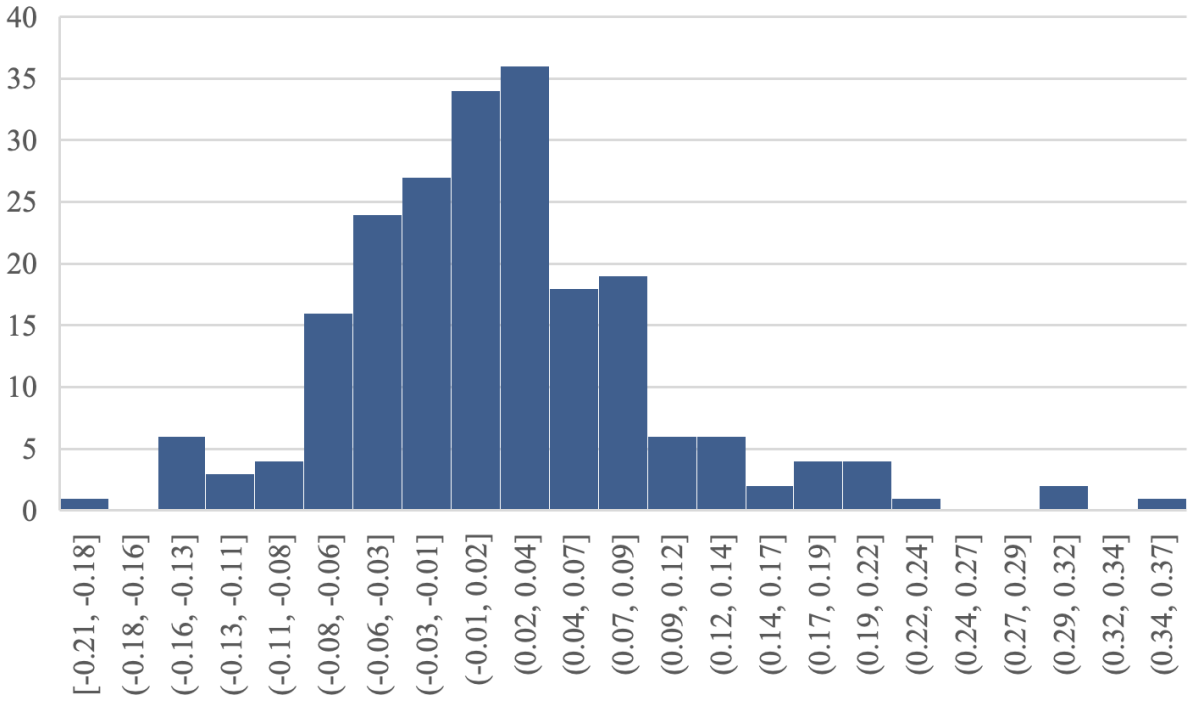


Figure 5.4: Empirical distribution)

Concerning liquidity, we have already mentioned that this paper estimates that the median spread for the FTSE STAR in the period January 2020-December 2024 stands at 104.4 bps, while the average spread is 160.9 bps, with an average daily turnover of €1,188,493.39. Although these data show that the segment has low liquidity, in principle they should not be able to eliminate abnormal returns completely, especially if the median spread, which is more robust to outliers, is used as a liquidity benchmark. However, for our analysis the relevance of these parameters could also be influenced by the heterogeneity of liquidity over the months or because the liquidity of securities in the pre-ex-date period could change significantly. This is why we also wanted to perform an analysis that calculates comprehensive average returns that take spreads into account and therefore consider the ask price as the purchase price and the bid price as the sale price. The results are shown in Table 5.5. As expected, returns have fallen significantly, and the returns of the strategy in the case of cutoffs of 0%, 0.25%, 1%, and 1.25% are even negative. Returns still grow with the dividend yield. Furthermore, the significance of the results has fallen sharply, and for many cutoffs, it is not possible to reject the hypothesis that average returns are not positive. The only cases in which it is possible to say that returns are significantly higher than 0 at the 5% level are those with cutoffs in the range [3.75%, 5%], while at the 10% level, they are those in the range [3%, 3.5%]. The returns in these ranges remain very high, ranging from 1.845% to 4.774% for the 15-day window. When annualized linearly for illustrative purposes only, these returns, which are shown in Table 5.6, translate into a range from 30.999% to 80.205%. However, attention should be paid to sample sizes. In summary, these data and liquidity estimates imply that the strategy is economically attractive only for dividend yields ranging from 3% to 5%.

Cutoff DivYield (%)	N events	AvgReturn 15d (%)	StdDev 15d (%)	Sharpe 15d
0	214	-0.255	8.434	-0.030
0.25	209	-0.201	8.476	-0.024
0.5	196	0.1405	8.348	0.017
0.75	175	0.0178	8.313	0.002
1	160	-0.015	8.348	-0.002
1.25	145	-0.129	8.192	-0.016
1.5	132	0.208	8.191	0.025
1.75	121	0.600	8.143	0.074
2	100	0.925	8.368	0.111
2.25	90	0.757	8.223	0.092
2.5	82	0.888	8.453	0.105
2.75	71	1.241	8.729	0.142
3	63	1.845*	9.001	0.205
3.25	51	2.075*	9.433	0.220
3.5	45	2.351*	9.394	0.250
3.75	39	2.920**	9.865	0.296
4	35	3.056**	10.228	0.299
4.25	28	4.437**	10.343	0.429
4.5	23	4.774**	10.417	0.458
4.75	20	3.970**	9.516	0.417
5	18	4.299**	9.979	0.431
5.25	14	4.002	11.228	0.356
5.5	14	4.002	11.228	0.356

Table 5.5: Returns summary (15 days window, considering Bid-Ask spreads)

Note: *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$

Cutoff DivYield (%)	N events	AvgReturn (lin. ann.,%)	AvgReturn (comp. ann.,%)	StdDev 15d (ann.,%)	Sharpe (ann.)
0	214	-4.295	-4.210	34.569	-0.124
0.25	209	-3.387	-3.334	34.742	-0.097
0.5	196	2.361	2.387	34.218	0.069
0.75	175	0.299	0.300	34.073	0.009
1	160	-0.261	-0.260	34.215	-0.008
1.25	145	-2.177	-2.155	33.577	-0.065
1.5	132	3.509	3.567	33.575	0.104
1.75	121	10.081	10.573	33.377	0.302
2	100	15.557	16.748	34.299	0.454
2.25	90	12.722	13.512	33.703	0.377
2.5	82	14.924	16.019	34.647	0.431
2.75	71	20.866	23.044	35.780	0.583
3	63	30.999	35.956	36.894	0.840
3.25	51	34.865	41.210	38.665	0.902
3.5	45	39.489	47.746	38.505	1.026
3.75	39	49.062	62.188	40.436	1.213
4	35	51.339	65.816	41.921	1.225
4.25	28	74.550	107.390	42.394	1.758
4.5	23	80.205	118.910	42.697	1.878
4.75	20	66.699	92.340	39.004	1.710
5	18	72.223	102.817	40.903	1.766
5.25	14	67.238	93.338	46.019	1.461
5.5	14	67.238	93.338	46.019	1.461

Table 5.6: Returns summary (annualized, considering Bid-Ask spreads)

5.2 Testing of multi-factor asset pricing models

After statistically testing whether it is actually possible to achieve positive returns in the period preceding the ex-date, it is necessary to verify the existence of the anomaly using the models already mentioned in the previous chapter. The results for the 15-day time window are shown in Table 5.7, while Table 5.8 shows the annualized alphas with the alphas expressed as percentages. In Table 5.8 the significance levels are not shown, because the sole purpose of showing annualized alphas is to provide alpha values to make them easily comparable with the necessary precautions.

By regressing the cumulative returns over the 15 days prior to the ex-date on the corresponding cumulative factors (CAPM, FF3, C4F, FF5), we obtain significantly positive intercepts for almost all dividend yield cutoffs. The window alpha grows monotonically as the yield increases ($\sim 1\%$ at the 0% cutoff, up to 3-5% for cutoffs $\geq 3\%$) and is robust to

the inclusion of profitability, investment, and momentum factors. This indicates that the pre-ex-date run-up is not explained by standard systematic risk alone and is more intense on high dividend yield names, in line with assumptions of price pressure and limits to arbitrage in a relatively illiquid segment. Statistical power is reduced at higher cutoffs due to smaller sample sizes, but the direction of the effect remains unchanged.

In several buckets, the estimated alpha increases and becomes more significant when moving from CAPM to multifactor models (up to FF5). This is consistent with two known mechanisms:

(i) correction of omitted variable bias – expected returns associated with tilt on size/value/momentum/profitability/investment are explicitly modeled; the benchmark rises, and the residual (intercept) better reflects the "pre-ex-date" premium;

(ii) reduction in residual variance - the additional factors explain more cross-section dispersion between events, compressing the standard error of the intercept; with the point estimate unchanged (or slightly increased), the t-stat rises and significance improves.

Cutoff DivYield (%)	N events	CAPM (%)	FF3 (%)	C4F (%)	FF5 (%)
0.00	214	0.96**	0.54	0.55	1.20**
0.25	209	0.94**	0.59	0.6	1.28***
0.50	196	1.26***	0.94**	0.95**	1.82***
0.75	175	1.34***	0.92**	0.92**	1.80***
1.00	160	1.38***	1.00**	1.03**	1.87***
1.25	145	1.45***	1.15**	1.22**	2.06***
1.50	132	1.69***	1.40***	1.47***	2.36***
1.75	121	1.91***	1.67***	1.72***	2.73***
2.00	100	2.26***	1.99***	2.00***	3.14***
2.25	90	2.16***	1.70***	1.70***	2.70***
2.50	82	2.17***	1.80***	1.76***	2.84***
2.75	71	2.23***	2.08***	1.99***	3.61***
3.00	63	2.73***	2.58***	2.47***	4.08***
3.25	51	3.04***	3.08***	2.99***	4.62***
3.50	45	2.50***	2.30**	2.85***	3.84***
3.75	39	3.04***	2.82***	3.51***	3.95***
4.00	35	2.58**	2.41**	3.17***	3.54***
4.25	28	3.09**	2.72*	4.14***	4.09***
4.50	23	2.88**	2.83**	3.78**	4.30***
4.75	20	2.45*	2.75*	3.53**	4.11***
5.00	18	2.97**	3.60**	3.66**	5.30***
5.25	14	1.66	2.26	2.31	4.64**
5.50	14	1.66	2.26	2.31	4.64**

Table 5.7: Alphas by model (15 days window)

Note: *** $p \leq 0.01$, ** $p \leq 0.05$, * $p \leq 0.10$

Cutoff DivYield (%)	N events	CAPM (%)	FF3 (%)	C4F (%)	FF5 (%)
0.00	214	16.128	9.072	9.240	20.160
0.25	209	15.792	9.912	10.080	21.504
0.50	196	21.168	15.792	15.960	30.576
0.75	175	22.512	15.456	15.456	30.240
1.00	160	23.184	16.800	17.304	31.416
1.25	145	24.360	19.320	20.496	34.608
1.50	132	28.392	23.520	24.696	39.648
1.75	121	32.088	28.056	28.896	45.864
2.00	100	37.968	33.432	33.600	52.752
2.25	90	36.288	28.560	28.560	45.360
2.50	82	36.456	30.240	29.568	47.712
2.75	71	37.464	34.944	33.432	60.648
3.00	63	45.864	43.344	41.496	68.544
3.25	51	51.072	51.744	50.232	77.616
3.50	45	42.000	38.640	47.880	64.512
3.75	39	51.072	47.376	58.968	66.360
4.00	35	43.344	40.488	53.256	59.472
4.25	28	51.912	45.696	69.552	68.712
4.50	23	48.384	47.544	63.504	72.240
4.75	20	41.160	46.200	59.304	69.048
5.00	18	49.896	60.480	61.488	89.040
5.25	14	27.888	37.968	38.808	77.952
5.50	14	27.888	37.968	38.808	77.952

Table 5.8: Alphas by model (annualized)

5.3 Returns event study

The final analysis concerns the analysis of average price returns around the ex-date and the verification of a hypothetical post-ex-date reversal. Figure 5.9 and Figure 5.10 show the average price returns per day and the cumulative trend of average price returns, respectively. In the event time profile of daily returns, there is a clear pre-ex-date run-up concentrated in the last week: 8 out of the 14 days before the ex-date have high and positive returns with a peak of 0.6% at t = -1, while 6 out of the 14 pre-ex-date days have slightly negative returns. The cumulative returns figure confirms that the gain builds up progressively until t = -1. At the ex-date (t = 0), there is a sharp drop (-2.2%), very close to the average net dividend yield of the sample ($\sim$ 2.4%), but still lower: this is exactly what is expected from the "ex-dividend" adjustment mechanism (Elton and Gruber 1970), especially in Italy, where uniform taxation on dividends and capital gains theoretically makes the price-drop ratio close to 1. After the ex-date, a partial mean

reversion emerges: the following days recover part of the decline and the cumulative values return to around zero within t = 10, a sign that a component of the run-up was transitory (price pressure/temporary inelasticity of demand) and is reabsorbed when the "dividend-seeking" demand disappears.

If we only consider events with a dividend yield $\geq 3.75\%$ (calculated using the share price at t-15), the analysis yields even more interesting results (Figure 5.11 & Figure 5.12). It was decided to analyze the 3.75% level because this level is among the levels with the highest 1% significant alphas and among the levels with the highest 5% significant returns (net of bid/ask spreads). The drop on the ex-date is -5.37%, slightly lower but in line with the average net dividend yield (5.52%) for shares with a dividend yield $\geq$ 3.75%, exactly as in the case of the complete sample. This further reinforces Elton and Gruber's theory regarding the "ex-dividend" adjustment mechanism, which should be in line with the net dividend yield. Unlike the complete sample, after the ex-date there is a prolonged negative drift, which at t = 13 reaches as much as -4.9%. In the pre-ex-date period, cumulative returns are almost strictly monotonically increasing, while in the post-ex-date period they are almost strictly monotonically decreasing. This further reinforces the theory that a component of the run-up was transitory (price pressure/temporary inelasticity of demand) and is reabsorbed when the "dividend-seeking" demand disappears. Operationally, these patterns motivate our choice to close the window at t = -1: holding the stock through the ex-date reduces the gain built up before. Although the average net dividend slightly exceeds the price drop on the ex-date, the total return benefit of 1 day is marginal (and its present value is only a few basis points and could even be negative depending on the discount rate used and the distance of the dividend payment from the ex-date), while the post-ex risk and costs are predominant. Closing at t = -1 therefore isolates the run-up premium linked to price pressure and arbitrage limits, which is the subject of the anomaly analyzed.

In economic terms, the combination of run-up, drop on the ex-date, and partial reversal is consistent with the interpretation of demand pressure on high-yield stocks in a relatively illiquid segment (STAR), with market makers/LPs and retail investors

accentuating crowding in the final days. The fact that this evidence becomes stronger as the dividend yield cutoff increases is perfectly in line with this interpretation.

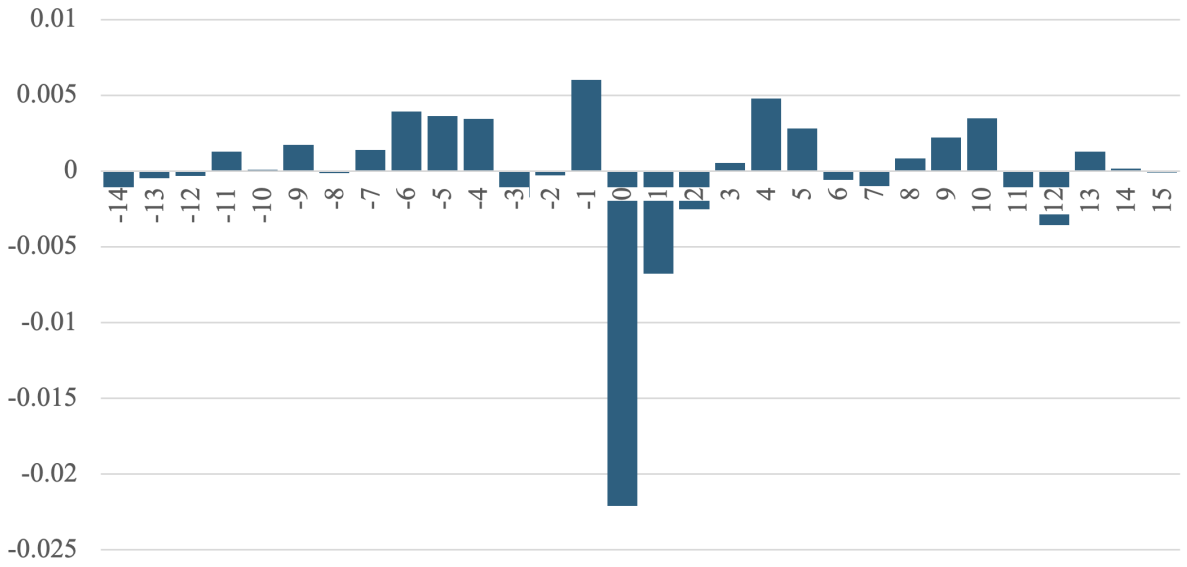


Figure 5.9: Daily price returns

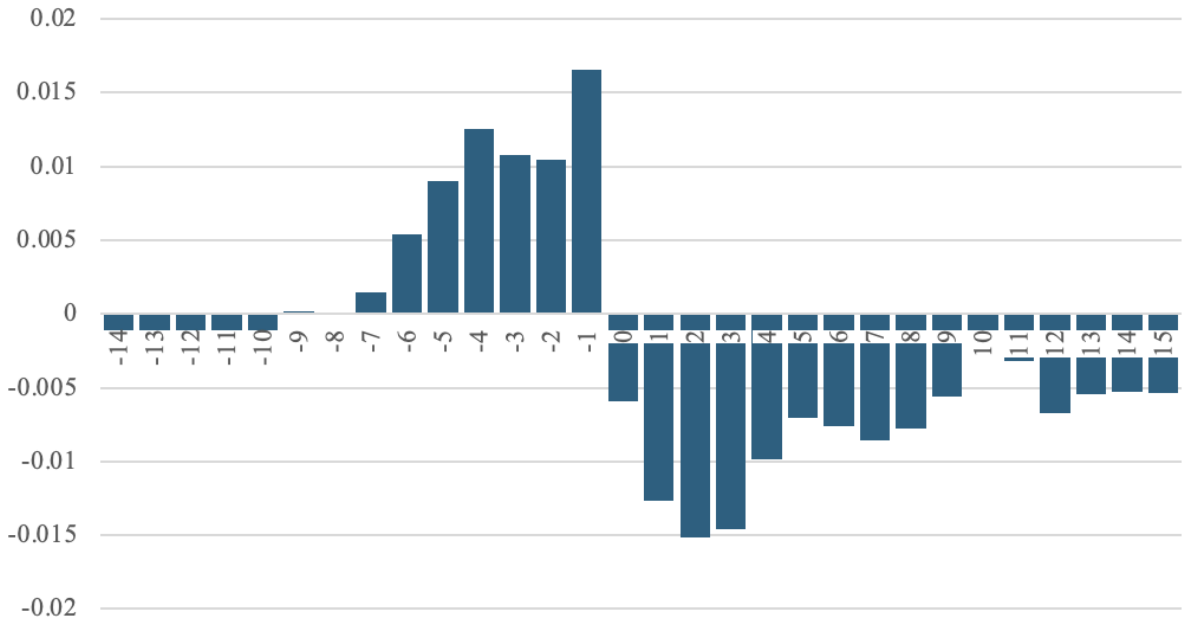


Figure 5.10: Cumulative daily price returns

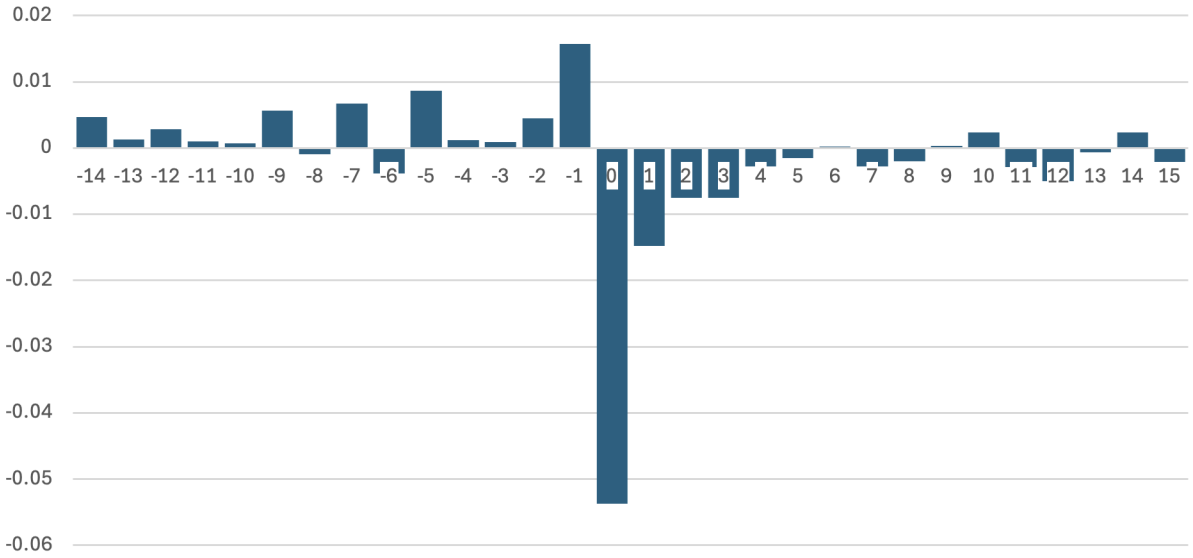


Figure 5.11: Daily price returns (for events with a dividend yield $\geq 3.75\%$)

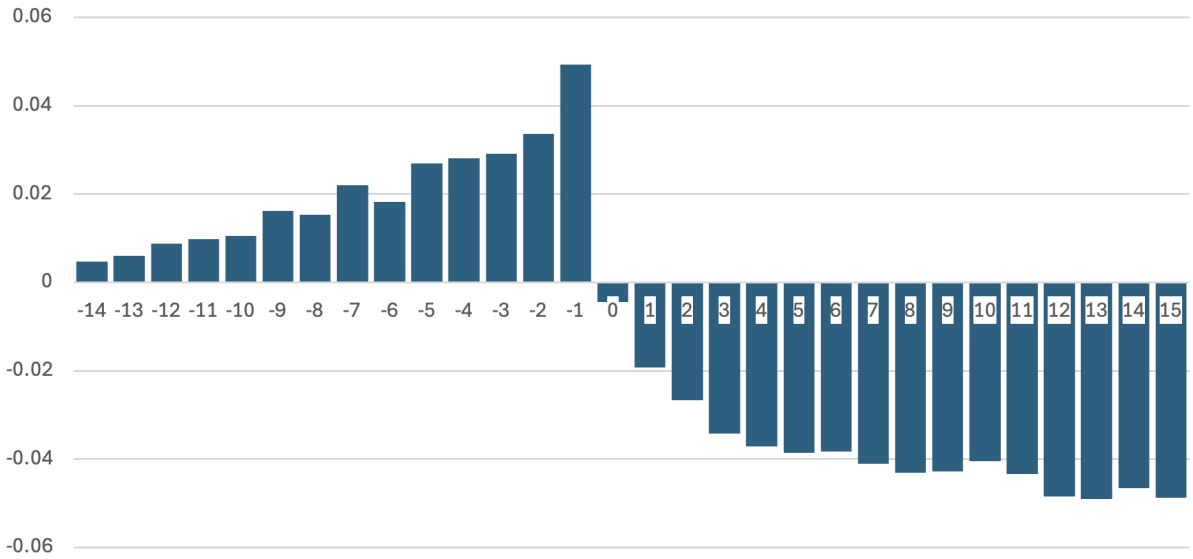


Figure 5.12: Cumulative daily price returns (for events with a dividend yield $\geq 3.75\%$)

6. Conclusion

This paper documents, for the FTSE STAR segment, a price run-up in the 15 days preceding the ex-date and positive and statistically significant alphas when window returns are regressed on cumulative factors (CAPM, FF3, C4F, FF5). The intensity of the anomaly increases with dividend yield: moving from cutoffs close to 0% to cutoffs $\geq$ 3%, the window alpha increases from around 1% to 3-5% and remains significant even in specific multifactorial models. If bid-ask spreads are also taken into account, returns decline but remain positive overall, especially as dividend yields increase, although they become less significant, particularly for lower dividend yields. The event study shows how the returns gradually accumulate until the day before the ex-date and then drop markedly at the ex-date, in line with the ex-dividend mechanism. Overall, the results are consistent with the hypothesis of price pressure fueled by dividend-seeking demand in a context of limited liquidity and arbitrage constraints. The most consistent theories with the FTSE STAR case are those related to catering and signaling. Meanwhile, clientele effects linked to tax reasons are not particularly consistent with the Italian context due to the equal tax treatment of dividends and capital gains.

In terms of methodology, evidence that significance often increases when moving from CAPM to FF5 is compatible with (i) correction of the bias of the omitted variables (profitability, investment, momentum), which leaves a 'cleaner' residual of the pre-ex-date premium, and (ii) reduction of the residual variance, which lowers the standard error of the intercept and increases the t-stat. This interpretation fits into the literature on Dividend Month Premium and price pressure (Hartzmark and Solomon 2013) and, for the Italian market profile, is consistent with the results on clientele / dividend preference, signaling and arbitrage limits.

6.1 Economic implications

For investors, the pattern suggests that the premium is more pronounced on high-yield names but less scalable: high-cutoff buckets have small samples and are found on more illiquid securities. For the market, the anomaly is indicative of demand inelasticity and how improvements in microstructure (spreads, market making, depth) can mitigate its magnitude.

6.2 Limitations

1) Small samples at high cutoffs lower the statistical power.

2) The 2020–2024 horizon is a period characterized by several market shocks, such as those caused by COVID-19 or the Russian-Ukrainian conflict, and the results are sensitive to the period.

3) Annualizations are illustrative only (windows that are not i.i.d. repeatable).

4) Transaction costs: the main analysis does not net them; evidence with bid-ask shows that spreads can materially reduce tradable profitability while leaving the price anomaly intact.

6.3 Contribution

We show that in the FTSE STAR, Italy's mid-small cap market, which is fiscally neutral between dividends and capital gains, pre-ex-date premiums are present and robust in multifactor models, grow with yield, and are accompanied by a clear run-up/drop/reversal profile. In a context where international literature for Italy is mixed, the focus on STAR (smaller and less liquid than the main index) helps to reconcile theory and evidence.

6.4 Future research

In the future, there are various topics or aspects related to this paper that can be explored in greater depth, such as: 1) extending the time/cross-section sample; 2) extending the analysis of the FTSE STAR segment using the dividend prediction methodology used by Hartzmark and Solomon; 3) analyzing liquidity and ownership subgroups; 4) extending the analysis to illiquid small- and mid-cap segments in other countries for comparison purposes; 5) extending the analysis to the FTSE MIB and the entire Italian market. 6) testing the robustness of abnormal returns to liquidity factors.

6.5 Final takeaway

In the STAR segment, the dividend run-up exists, is not explained by standard factors, and is stronger where yields are high and liquidity is low. For empirical finance, it is a textbook case of inelastic demand; for the market, it is a reminder that microstructure and investor base composition matter at least as much as systematic risk.

References

Ainsworth, Andrew and Maximilian Nicholson (2015). "Can dividend schedules predict abnormal returns? International evidence". In: *Financial Markets & Corporate Governance Conference*. URL: https://acfr.aut.ac.nz/__data/assets/pdf_file/0003/29712/581064-A-Ainsworth-DividendSchedules_AFM.pdf.

Avio S.p.A. (August 15, 2025). *Who We Are*. URL: https://www.avio.com/index.php/who-we-are (visited on 08/15/2025).

Baker, Malcolm and Jeffrey Wurgler (2004). "A catering theory of dividends". In: *The Journal of Finance* 59.3, pp. 1125–1165. DOI: 10.1111/j.1540-6261.2004.00658.x.

Beaver, William H. (1968). "The information content of annual earnings announcements". In: *Journal of Accounting Research* 6, pp. 67–92. DOI: 10.2307/2490070.

Becker, Bo, Zoran Ivković, and Scott Weisbenner (2009). *Local dividend clienteles*. NBER Working Paper 15196. National Bureau of Economic Research. DOI: 10.3386/w15196.

Bhattacharya, Sudipto (1979). "Imperfect information, dividend policy, and "the bird in the hand" fallacy". In: *The Bell Journal of Economics* 10.1, pp. 259–270. DOI: 10.2307/3003330.

Black, Fischer, Michael C. Jensen, and Myron Scholes (1972). "The capital asset pricing model: Some empirical tests". In: *Studies in the Theory of Capital Markets*. Ed. by Michael C. Jensen. New York: Praeger, pp. 79–121. URL: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=908569.

Borsa Italiana S.p.A. (August 15, 2025a). *Homepage*. URL: https://www.borsaitaliana.it/homepage/homepage.htm (visited on 08/15/2025).

Borsa Italiana S.p.A. (August 15, 2025b). *Regolamenti : Guide*. URL: https://www.borsaitaliana.it/borsaitaliana/regolamenti/guide/guide.en.htm (visited on 08/15/2025).

Carhart, Mark M. (1997). "On persistence in mutual fund performance". In: *The Journal of Finance* 52.1, pp. 57–82. DOI: 10.1111/j.1540-6261.1997.tb03808.x.

Chae, Young Kyung and Hyun-Koo Koo (2020). "Dividend-month premium in the Korean stock market". In: *Journal of Derivatives and Quantitative Studies* 28.3, pp. 193–213. DOI: 10.1108/JDQS-04-2020-0006.

Chai, Daniel, Viet Do, and Maria Strydom (2025). *The dividend month premium uncovered. Working paper*. Department of Banking and Finance, Monash University. URL: https://acfr.aut.ac.nz/__data/assets/pdf_file/0004/105493/90285-Dividend-premium-Uncovered-.pdf (visited on 08/15/2025).

CONSOB – Commissione Nazionale per le Società e la Borsa (2024). *Report on the Financial Investments of Italian Households 2024*. CONSOB. URL: https://www.consob.it/web/consob-and-its-activities/abs-rf/-/asset_publisher/45Riev4mErvY/content/rf2024/718268 (visited on 08/15/2025).

— (2025). *Trends and Challenges for the Italian Financial Sector: Economic Report 2024*. URL: https://www.consob.it/documents/d/area-pubblica/economicreport2024 (visited on 08/15/2025).

Elton, Edwin J. and Martin J. Gruber (1970). "Marginal stockholder tax rates and the clientele effect". In: *The Review of Economics and Statistics* 52.1, pp. 68–74. DOI: 10.2307/1927599.

Fama, Eugene F. (1970). "Efficient capital markets: A review of theory and empirical work". In: *The Journal of Finance* 25.2, pp. 383–417. DOI: 10.2307/2325486.

Fama, Eugene F. and Kenneth R. French (1993). "Common risk factors in the returns on stocks and bonds". In: *Journal of Financial Economics* 33.1, pp. 3–56. DOI: 10.1016/0304-405X(93)90023-5.

Fama, Eugene F. and Kenneth R. French (2012). "Size, value, and momentum in international stock returns". In: *Journal of Financial Economics* 105.3, pp. 457–472. DOI: 10.1016/j.jfineco.2012.05.011.

— (2015). "A five-factor asset pricing model". In: *Journal of Financial Economics* 116.1, pp. 1–22. DOI: 10.1016/j.jfineco.2014.10.010.

— (2017). "International tests of a five-factor asset pricing model". In: *Journal of Financial Economics* 123.3, pp. 441–463. DOI: 10.1016/j.jfineco.2016.11.004.

Fama, Eugene F. and James D. MacBeth (1973). "Risk, return, and equilibrium: Empirical tests". In: *Journal of Political Economy* 81.3, pp. 607–636. DOI: 10.1086/260061.

Foye, James, Dušan Mramor, and Marko Pahor (March 2013). "A Respecified Fama–French Three-Factor Model for the New European Union Member States". In: *Journal of International Financial Management & Accounting* 24.1, pp. 3–25. DOI: 10.1111/jifm.12005.

Frazzini, Andrea and Owen A. Lamont (2008). "Dumb money: Mutual fund flows and the cross-section of stock returns". In: *Journal of Financial Economics* 88.2, pp. 299–322. DOI: 10.1016/j.jfineco.2007.07.011.

Graham, John R. and Alok Kumar (2006). "Do dividend clienteles exist? Evidence on dividend preferences of retail investors". In: *The Journal of Finance* 61.3, pp. 1305–1336. DOI: 10.1111/j.1540-6261.2006.00873.x.

Griffin, John M. (2002). "Are the Fama and French factors global or country specific?" In: *Review of Financial Studies* 15.3, pp. 783–803. URL: https://dx.doi.org/10.2139/ssrn.262647.

Hartzmark, Samuel M. and David H. Solomon (2013). "The dividend month premium". In: *Journal of Financial Economics* 109.3, pp. 640–660. DOI: 10.1016/j.jfineco.2013.03.005.

Heston, Steven L. and Ronnie Sadka (2004). "Seasonality in the Cross-Section of Expected Stock Returns". AFA 2006 Boston Meetings Paper. DOI: 10.2139/ssrn.687022.

Jensen, Michael C. (1968). "The performance of mutual funds in the period 1945–1964". In: *The Journal of Finance* 23.2, pp. 389–416. DOI: 10.1111/j.1540-6261.1968.tb00815.x.

Li, Wei and Erik Lie (2006). "Dividend changes and catering incentives". In: *Journal of Financial Economics* 80.2, pp. 293–308. DOI: 10.1016/j.jfineco.2005.03.005.

Mancinelli, Laura and Aydin Ozkan (2006). "Ownership structure and dividend policy: Evidence from Italian firms". In: *The European Journal of Finance* 12.3, pp. 265–282. DOI: 10.1080/13518470500249365.

Markowitz, Harry (1952). "Portfolio selection". In: *The Journal of Finance* 7.1, pp. 77–91. DOI: 10.2307/2975974.

— (1959). *Portfolio Selection: Efficient Diversification of Investments*. New York, NY: Wiley. URL: https://www.jstor.org/stable/j.ctt1bh4c8h.

Michaely, Roni, Jean-Luc Vila, and Jiang Wang (1996). "A model of trading volume with tax-induced heterogeneous investors". In: *Journal of Financial Intermediation* 5.3, pp. 340–371. DOI: 10.1006/jfin.1996.0015.

Miller, Merton H. and Kevin Rock (1985). "Dividend policy under asymmetric information". In: *The Journal of Finance* 40.4, pp. 1031–1051. DOI: 10.1111/j.1540-6261.1985.tb02362.x.

Neumann, John von and Oskar Morgenstern (1947). *Theory of Games and Economic Behavior*. 2nd ed. Princeton: Princeton University Press.

OECD/INFE (2023). *International Survey of Adult Financial Literacy*. Paris: OECD Publishing. DOI: 10.1787/56003a32-en.

PwC (September 5, 2025). *Italy: Corporate: Withholding Taxes*. URL: https://taxsummaries.pwc.com/italy/corporate/withholding-taxes (visited on 09/17/2025).

Sabaf S.p.A. (August 15, 2025). *About Us*. URL: https://www.sabaf.it/en/our-company (visited on 08/15/2025).

Sharpe, William F. (1964). "Capital asset prices: A theory of market equilibrium under conditions of risk". In: *The Journal of Finance* 19.3, pp. 425–442. DOI: 10.1111/j.1540-6261.1964.tb02865.x.

Shefrin, Hersh and Meir Statman (1984). "Explaining investor preference for cash dividends". In: *Journal of Financial Economics* 13.2, pp. 253–282. DOI: 10.1016/0304-405X(84)90025-4.

Spencer Stuart (2024). *2024 S&P 500 Industry Sector Comparison Chart*. URL: https://www.spencerstuart.com/research-and-insight/sector-comparison-chart (visited on 08/15/2025).

Virtu (2024). *2024 US Market Structure Report*. Virtu Financial.

Wang, Jiang (1996). "The term structure of interest rates in a pure exchange economy with heterogeneous investors". In: *Journal of Financial Economics* 41.1, pp. 75–110. DOI: 10.1016/0304-405X(95)00854-8.

Yahoo Finance (August 15, 2025). *Quote*. URL: https://it.finance.yahoo.com/quote/ (visited on 08/15/2025).